# BJT Collector-base Coupled Non-Inverting Schmitt Trigger

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## 1. Introduction

## 2. Schematics

## 3. Experimental Setup

### 3.1. Experiment 1: BJT Collector-base Coupled Non-Inverting Schmitt Trigger Characterization

In this experiment, we set up the circuit as shown in Figure 1 and measured the output voltage of the circuit. We swept the voltage from $V_{ee}$ (GND) to $V_{cc}$ and then back to $V_{ee}$. We are characterizing the Upper and Lower Threshold Voltages of the circuit (represented by $V_{UT}$ and $V_{LT}$ respectively) and then calculating the Hysteresis Voltage (represented by $V_H$) by subtracting the Upper Threshold Voltage from the Lower Threshold Voltage.

$$V_H = V_{UT} - V_{LT}$$

With the same data, we also extracted the Output High and Low Voltage (represented by $V_{OH}$ and $V_{OL}$ respectively) by looking at what the average high voltage was and what the average low voltage was.

### 3.2. Experiment 2: Schmitt Trigger

## 4. Results

## 5. Analysis

## 6. Discussion

## 7. Summary

## 8. Sources

- Schmitt Trigger Explained (Design of Inverting and Non-inverting Schmitt Trigger using Op-Amp) Video - https://youtu.be/5-ohKRWeod4?si=8VM2gfX6_NjWiwgg

## 9. notes

- Has to be a gap between R1 and R2 values and R3 and R4 values; R1/R2 must be smaller than R3/R4. - Output High is determined by voltage divider ratio betwene R2 and R4 where they are a volt div - When one turns one, it sucks current from the other base of other BJT and turns the other off at the same time

**Figure 1.** caption here